Exp:2 Learning and assignment of IP addresses manually to computers.

AIM: To understand the IP addressing scheme and manually assign static IP addresses to computers on a network.

Procedure:

1) In Windows: Open Networks and Sharing Centre → Change Adapter Setting → Right-click Local Area Connection → Properties → IPV4 → Set IP Manually.

2) Enter IP addresses: 192.168.1.10, Subnet Mask: 255255.255.0 (Default Gateway: 192.168.1.1.

3) Enter DNS Server: 8.8.8.8 (Google DNS), click Ok.

4) In Linux: Use command 'Sudo ip route add default via 192.168.1.1'.

5) For persistent configuration in Linux: edit /etc/network/interfaces or use nmcli.

6) Verify the configuration using 'ipconfig' or 'ip a'.

7) Ping the default gateway to confirm network connectivity

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3) Ping another computer on the same subnet to verify communication.

Result:

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Static IP addresses were successfully assigned to computers. Connectivity was verified using ping commands between hosts on the same subnet.